



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

CHEMISTRY
JC2 Preliminary Examination
Paper 4 Practical

9729/04
25 Aug 2017
2 hr 30 min

Candidates answer on the Question Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Give details of the practical shift and laboratory in the boxes provided.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Qualitative Analysis Notes are printed on pages 22 and 23.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part questions.

Shift
Laboratory

For Examiner's Use	
1	/14
2	/19
3	/12
4	/10
TOTAL	/ 55

This document consists of 22 printed pages and 2 blank pages.

Answer **all** the questions in the spaces provided.

- 1** You are to determine the concentration of aqueous copper(II) sulfate by titration. The concentration of Cu^{2+} ions in a solution can be found by reaction with an excess of aqueous iodide ions to produce iodine. The amount of iodine formed can be found by titration with thiosulfate ions, $\text{S}_2\text{O}_3^{2-}$.

*For
Examiner's
use*

FB 1 is aqueous copper(II) sulfate, CuSO_4 .

FB 2 is $0.100 \text{ mol dm}^{-3}$ sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$.

FB 3 is aqueous potassium iodide, KI .

starch indicator

(a) Method

- (i)**
1. Fill the burette with **FB 2**.
 2. Using a pipette, transfer 25.0 cm^3 of **FB 1** into a conical flask.
 3. Use an appropriate measuring cylinder, transfer 10 cm^3 of **FB 3** to the same conical flask.
 4. Titrate this mixture with **FB 2** until the colour of the mixture changes from brown to yellow-brown. An off-white precipitate will also be present in the flask throughout the titration.
 5. Add 5 drops of starch indicator.
 6. Continue the titration until the blue-black colour of the starch-iodine complex just disappears leaving the off-white precipitate.
 7. Record your titration results in the space provided below. Make certain that your recorded results show the precision of your working.
 8. Repeat points **1** to **7** as necessary until consistent results are obtained.

Results

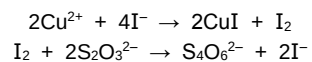
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[5]

- (ii)** From your titrations, obtain a suitable volume of **FB 2** to be used in your calculations. Show clearly how you obtained this volume.

volume of **FB 2** =[1]

- (b) (i) The equations for the formation of iodine and its reaction with thiosulfate ions are given below.



Calculate the amount of thiosulfate ions, $\text{S}_2\text{O}_3^{2-}$, present in the volume of **FB 2** in (a)(ii).

For
Examiner's
use

amount of $\text{S}_2\text{O}_3^{2-}$ =[1]

- (ii) Using the equations above, calculate the amount of Cu^{2+} ions present in 25.0 cm^3 of **FB 1**.

amount of Cu^{2+} =[1]

- (iii) Calculate the concentration, in mol dm^{-3} , of copper(II) sulfate in **FB 1**.

concentration of CuSO_4 =[1]

- (c) Two students repeated the experiment but each obtained different values for the concentration of CuSO_4 .

For
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use

The students each suggested possible improvements.

Student 1 suggested that a larger volume of potassium iodide, **FB 3**, should be added.
Student 2 suggested that the contents of the conical flask should be filtered before titration.

Comment on the effectiveness of **each** of these possible improvements.
Explain your answers.

Student 1

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Student 2

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[2]

For
Examiner's
use

- (d) The maximum error in each burette reading is $\pm 0.05 \text{ cm}^3$.
The maximum error in each titration is therefore $\pm 0.10 \text{ cm}^3$.
The 25.0 cm^3 pipette, used in this titration, is labelled with an error of $\pm 0.06 \text{ cm}^3$.

Explain why the maximum error when 25.0 cm^3 of solution is run from a pipette is only $\pm 0.06 \text{ cm}^3$ and not $\pm 0.12 \text{ cm}^3$.

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[1]

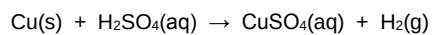
- (e) Considering the maximum errors from the pipette and burette used, calculate the total percentage error in one titration.

[2]

[Total: 14]

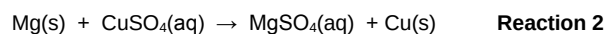
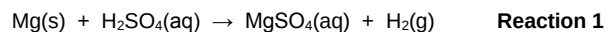
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- 2 You are to determine the enthalpy change of reaction, ΔH , for the reaction shown below.



For
Examiner's
use

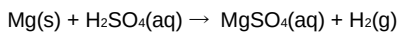
Since copper is an unreactive metal it does not react directly with dilute acids. You will therefore need to find the enthalpy change of reaction for two reactions that do occur. The equations for these two reactions are below.



You will carry out experiments to find the enthalpy changes for each of **Reaction 1** and **Reaction 2** and use these values to calculate the enthalpy change for the reaction of copper with sulfuric acid.

TURN OVER FOR EXPERIMENTAL METHOD

Determining the enthalpy change for Reaction 1



Reaction 1

For
Examiner's
use

(a) Method

FB 4 is 1.00 mol dm⁻³ sulfuric acid, H₂SO₄.

FB 5 is magnesium powder, Mg.

Read through the method before you start any practical work and prepare suitable tables for your results.

1. Weigh the stoppered weighing bottle containing **FB 5**. Record the mass.
2. Support the styrofoam cup in the 250cm³ beaker.
3. Use the measuring cylinder to transfer 25cm³ of **FB 4** into the styrofoam cup.
4. Measure the temperature of **FB 4** in the styrofoam cup and start the stop watch. Record this temperature as being the temperature at time = 0.
5. Measure, and record, the temperature of this **FB 4** every half minute for 2 minutes.
6. At time = 2.5 minutes, add **FB 5** to the acid and stir carefully to reduce acid spray.
***Caution: Inhalation of the fumes may cause choking.**
7. Measure the temperature of the mixture in the cup at time = 3 minutes and then every half minute up to time = 7 minutes.
8. Continue stirring occasionally throughout this time.
9. Weigh the weighing bottle that had contained **FB 5**. Record the mass.
10. Calculate and record the mass of **FB 5** added to the sulfuric acid.
11. Discard the used styrofoam cup.

Results

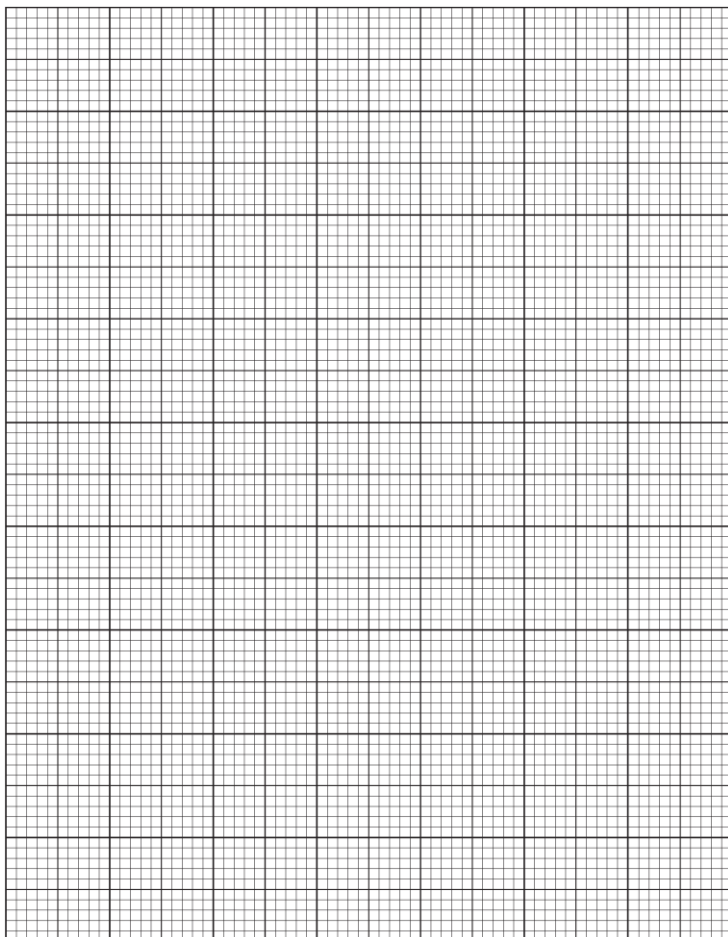
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[5]

- (b) (i) On the grid below plot a graph of temperature (y -axis) against time (x -axis).

[2]

For
Examiner's
use



- (ii) Complete the graph by inserting **two, straight** lines of best fit:

- one to show the temperature up to time = 2.5 minutes,
- one to show the temperature after time = 2.5 minutes.

[1]

- (iii) From your graph, use the two straight lines of best fit to calculate the change in temperature at time = 2.5 minutes.

temperature change =[1]

(c) Calculations

- (i) In the reaction in **(a)**, the sulfuric acid was in excess. Without carrying out any additional tests, what observation could you have made during your experiment to confirm this?

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[1]

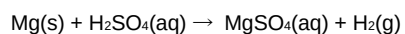
- (ii) Calculate the energy change that occurred during the reaction in **(a)**.

[Assume that 4.2 J is needed to raise the temperature of 1.0 cm³ of solution by 1.0°C.]

energy change =[1]

- (iii) Use your answer to **(ii)** to calculate the enthalpy change, in kJ mol⁻¹, for the reaction between sulfuric acid and magnesium.

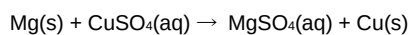
[Ar: Mg, 24.3]



Reaction 1

enthalpy change for **Reaction 1** =[1]

Determining the enthalpy change for Reaction 2



Reaction 2

For
Examiner's
use

(d) Method

FB 6 is 1.00 mol dm⁻³ copper(II) sulfate, CuSO₄.

FB 7 is magnesium powder, Mg.

Read through the method before you start any practical work and prepare suitable tables for your results.

1. Weigh the weighing bottle containing **FB 7**. Record the mass.
2. Support the styrofoam cup in the 250cm³ beaker.
3. Use the measuring cylinder to transfer 25cm³ of **FB 6** into the styrofoam cup.
4. Measure the temperature of **FB 6** in the styrofoam cup and record the temperature.
5. Add the **FB 7** to the **FB 6** in the cup and stir the mixture constantly.
6. Measure and record the maximum temperature reached during the reaction.
7. Calculate and record the maximum temperature change that occurred during the reaction.
8. Weigh the weighing bottle that had contained **FB 7**. Record the mass.
9. Calculate and record the mass of **FB 7** added to the copper(II) sulfate.
10. Empty the contents of the styrofoam cup into the bottle labelled **waste**.
11. Discard the used styrofoam cup.

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Results

I	
II	

[2]

(e) Calculations*For
Examiner's
use*

- (i) Show, using suitable calculations, that the copper(II) sulfate was in excess in the reaction.
[Ar : Mg, 24.3]

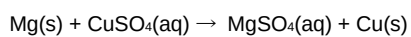
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[1]

- (ii) Hence, calculate the enthalpy change, in kJ mol^{-1} , for the reaction between magnesium and copper(II) sulfate.

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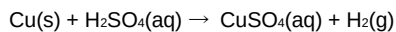
[Assume that 4.2 J is needed to raise the temperature of 1.0cm^3 of solution by 1.0°C .]

**Reaction 2**

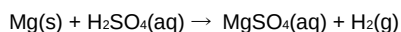
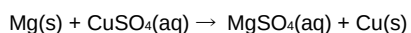
enthalpy change for **Reaction 2** =[1]

Enthalpy change for Reaction 3

Reaction 3 is shown below.

**Reaction 3**

- (f) Use your values for the enthalpy changes for **Reactions 1** and **2** to calculate the enthalpy change for **Reaction 3**.

**Reaction 1****Reaction 2**

Show clearly how you obtained your answer.

(If you were unable to calculate the enthalpy changes for **Reactions 1** and **2**, you should assume that the value for **Reaction 1** is -444 kJ mol^{-1} and that the value for **Reaction 2** is -504 kJ mol^{-1} . Note: these are not the correct values.)

enthalpy change for **Reaction 3** =[1]

- (g) (i) The method you used to determine the enthalpy change for **Reaction 1** was more accurate than the method you used to determine the enthalpy change for **Reaction 2**. Suggest why the method used for **Reaction 2** was less accurate. Explain your answer.

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[1]

- (ii) A student suggested that the accuracy of the method used for **Reaction 2** could be improved by using a larger volume of copper(II) sulfate. Is this a correct suggestion? Give a reason for your answer.

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[1]

[Total : 19]

For
Examiner's
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3 Qualitative Analysis

For
Examiner's
use

At each stage of any test, you are to record details of the following.

- Colour changes seen
- The formation of any precipitate
- The solubility of such precipitates in an excess of the reagent added

Where gases are released, they should be identified by a test, described in the appropriate place in your observations.

You should indicate clearly at what stage in a test a change occurs.

No additional tests for ions present should be attempted.

FB 8 is a solution containing 2 cations and 1 anion.

(a)

Test	Procedure	Observation
1	To 1 cm depth of FB 8 in a test tube, add aqueous ammonia until it is in excess.	
2	To 1 cm depth of FB 8 in a boiling tube , add aqueous sodium hydroxide until it is in excess. Then heat the boiling tube gently with care. Keep the mixture for Test 3 .	
3	To the mixture from Test 2 , add 1 cm depth of aqueous hydrogen peroxide. Shake the boiling tube gently.	
4	To 1 cm depth of FB 8 in a test tube, add aqueous barium chloride, followed by dilute hydrochloric acid.	

[4]

- (b) Using your observations above, identify the ions in **FB 8**. Give evidence to support your conclusion.

For
Examiner's
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[3]

- (c) What type of reaction has taken place in **Test 3**? Explain your answer.

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[1]

- (d) **FB 9** is an aqueous solution, containing one cation and one anion. Carry out the following tests in test-tubes. Complete the table by recording your observations.

For
Examiner's
use

Test	Procedure	Observations
5	To 1 cm depth of FB 9 , add aqueous sodium hydroxide until it is in excess.	
6	To 1 cm depth of FB 9 , add a piece of magnesium ribbon.	
7	To 1 cm depth of FB 9 , add 2 drops of aqueous silver nitrate, followed by aqueous ammonia until in excess.	

[2]

- (e) Using your observations above, identify the ions in **FB 9**. Give evidence to support your conclusion.

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[2]

[Total: 12]

- 4 Many transition metal complex ions are coloured. It is possible to use this property to determine the concentration of a solution of a coloured ion. A few cm³ of the solution is placed inside a *spectrometer*.

For
Examiner's
use

A *spectrometer* measures the amount of light that is absorbed when a specific wavelength of visible light is shone through a coloured solution. It does this by comparing the amount of light passing through the sample with the amount of light passing through the pure solvent. The amount of light absorbed is expressed as an absorbance value. The more concentrated the solution, the higher the absorbance value.

Beer-Lambert's Law states that the absorbance values, A , is directly proportional to the concentration of absorbing species, c , as shown below. The general Beer-Lambert's Law is usually written as,

$$A = \epsilon cl$$

Where ϵ is the molar extinction coefficient and l is the path length, which is usually 1.0cm.

This equation can be used to calculate the absorbance value when the concentration of copper(II) ions is known.

You may assume that you are provided with the following in the subsequent parts of the question.

- **FB 1** from Question 1
 - solid hydrated copper(II) sulfate, CuSO₄·5H₂O ($M_r = 249.6$);
 - access to a spectrometer and instructions for its use;
 - graph paper;
 - the apparatus and chemicals normally found in a school or college laboratory.
- (a) The spectrometer is set to use the wavelength of light that is absorbed most strongly by the copper(II) ions.

Suggest a colour in the visible spectrum from which a suitable wavelength of light might be chosen. Explain your answer.

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[1]

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- (b) Spectrometry can be used to determine the concentration of a solution of aqueous copper(II) sulfate, **FB 1**. A series of known, but different, concentrations of copper(II) sulfate is prepared. A spectrometer is used to measure the absorbance of each solution.

According to Beer-Lambert's Law, a graph of absorbance against concentration is then plotted. This graph is known as *calibration* line.

The experiment is then repeated using a solution of unknown concentration. By comparing the absorbance of this solution with the calibration line, the concentration of copper(II) ions in the unknown solution can be determined.

Propose a simple plan on how you would prepare

- a 100.0 cm³ of 1.00 mol dm⁻³ standard aqueous copper(II) sulfate;
- a suitable range of diluted solutions of accurate concentrations

You are to show detailed calculations and suitable tables (where appropriate) in your answer.

- (i) a 100.0 cm³ of 1.00 mol dm⁻³ standard aqueous copper(II) sulfate;

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- (ii) a suitable range of diluted solutions of accurate concentrations, keeping the total volume of each solution constant at 20.00 cm^3 .

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[3]

- Describe a plan to determine the concentration of **FB 1**.

- calculation of the absorbance value for each copper(II) sulfate solution prepared in **(b)**, given ϵ of CuSO_4 is $2.81 \text{ mol}^{-1} \text{ dm}^3 \text{ cm}^{-1}$;
- a sketch of the calibration line you would expect to obtain;
- a brief outline of how the results would be obtained.
- how the calibration line would be used to determine the concentration of copper(II) ions in **FB 1**.

[illegible]

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- (d) Given that the absorbance for **FB 1** is 0.286, calculate the concentration of **FB 1**.

For
Examiner's
use

concentration of **FB 1** =[1]

[Total : 10]

Qualitative Analysis Notes

[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	reaction with	
	NaOH(aq)	NH ₃ (aq)
aluminium, Al ³⁺ (aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH ₄ ⁺ (aq)	ammonia produced on heating	–
barium, Ba ²⁺ (aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca ²⁺ (aq)	white ppt. with high [Ca ²⁺ (aq)]	no ppt.
chromium(III), Cr ³⁺ (aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu ²⁺ (aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe ²⁺ (aq)	green ppt., turning brown on contact with air insoluble in excess	green ppt., turning brown on contact with air insoluble in excess
iron(III), Fe ³⁺ (aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg ²⁺ (aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn ²⁺ (aq)	off-white ppt., rapidly turning brown on contact with air insoluble in excess	off-white ppt., rapidly turning brown on contact with air insoluble in excess
zinc, Zn ²⁺ (aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

<i>ion</i>	<i>reaction</i>
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $\text{Cl}^-(\text{aq})$	gives white ppt. with $\text{Ag}^+(\text{aq})$ (soluble in $\text{NH}_3(\text{aq})$)
bromide, $\text{Br}^-(\text{aq})$	gives pale cream ppt. with $\text{Ag}^+(\text{aq})$ (partially soluble in $\text{NH}_3(\text{aq})$)
iodide, $\text{I}^-(\text{aq})$	gives yellow ppt. with $\text{Ag}^+(\text{aq})$ (insoluble in $\text{NH}_3(\text{aq})$)
nitrate, $\text{NO}_3^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil
nitrite, $\text{NO}_2^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil; NO liberated by dilute acids (colourless NO $\rightarrow$ (pale) brown NO_2 in air)
sulfate, $\text{SO}_4^{2-}(\text{aq})$	gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (insoluble in excess dilute strong acids)
sulfite, $\text{SO}_3^{2-}(\text{aq})$	SO_2 liberated with dilute acids; gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (soluble in dilute strong acids)

(c) Tests for gases

<i>gas</i>	<i>test and test result</i>
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

<i>halogen</i>	<i>colour of element</i>	<i>colour in aqueous solution</i>	<i>colour in hexane</i>
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple

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